

Rutgers School of Engineering + NJ Space Grant + JJ Slade



Overview

- SoE JJ Slade + NJ Space Grant
- Projects with NJ High School students encouraged to present at a national conference in the URTC at MIT.
- Spring Planning, Summer Execution
- Results
 - 51 High Schools Students
 - Over 45 distinct High Schools
 - 28/40 State Legislative Districts
 - 18/21 Counties
 - Projects served earned an average rating of 9.13/10

Scholarship Recipients



Krishiv A Patel



Hugh Brennan



Arnav Singh



Emily A. Li



Dominic
Plesniarski



Dario Gosevski



Shreya Srikanth



Jonah Simone

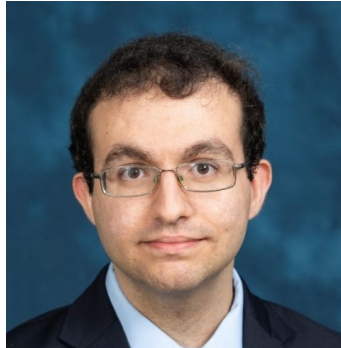


Marc Oliu

Faculty



Dr. Edward DeMauro



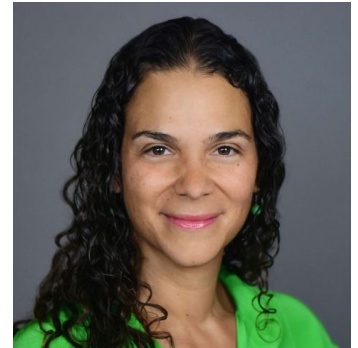
Dr. Yossi Cohen



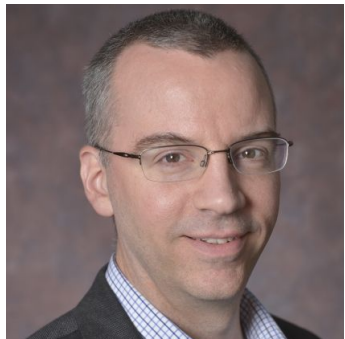
Dr. Stephen D. Tse



Dr. Jonathan P. Singer



Dr. Maribel Vazquez



Dr. Mark Pierce



Dr. Onur Bilgen



Dr. Laurent Burlion



Dr. Jason Yang



Dr. Prosenjit Bagchi



Krishiv Amit Patel

JJ Slade '26-'27

Senior, B.S. Aerospace Engineering, Energy
Concentration, Mathematics Minor

Our Team

Research Mentor:

Krishiv A. Patel

B.S. Aerospace Engineering, Energy
Concentration, Mathematics Minor

Residential Teaching Assistant:

Heather Fenton

B.S. Mechanical Engineering

Research Advisor:

Dr. Edward DeMauro

Department of Mechanical &
Aerospace Engineering



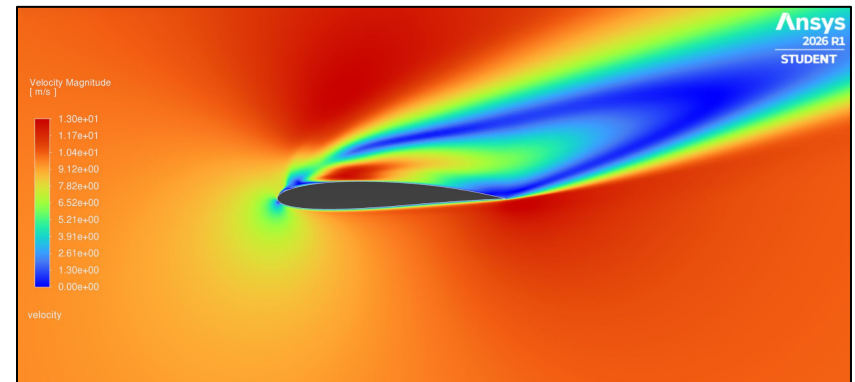
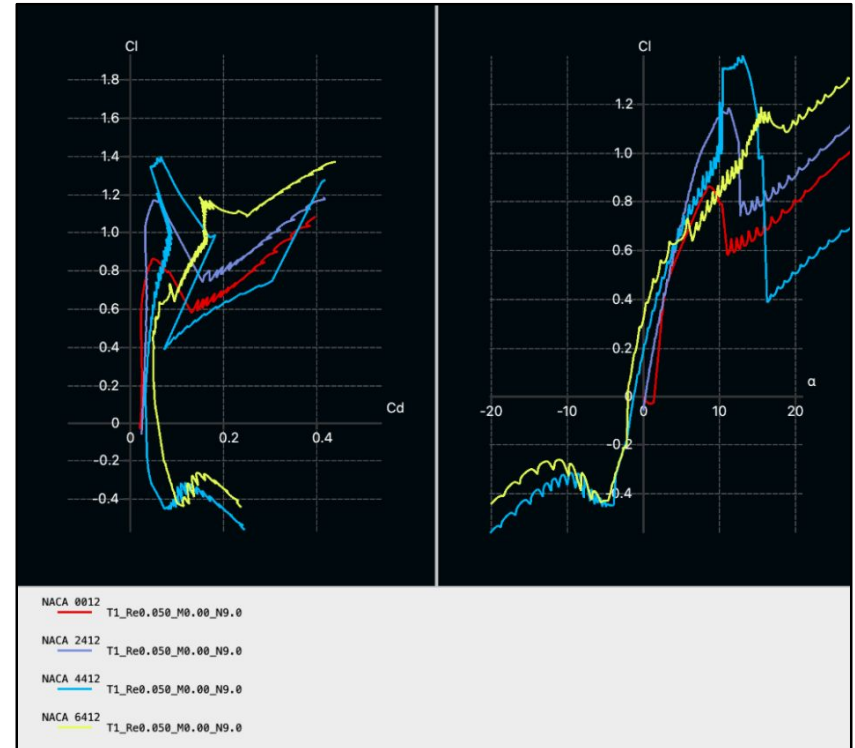
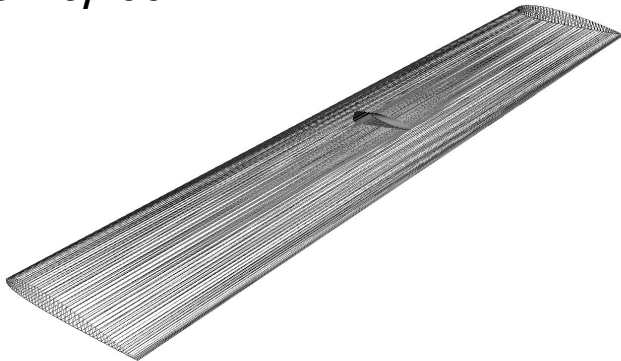
Scholars:

Parker Sherwood		Mainland Regional High School
Samuel Tobachnick		Wallkill Valley Regional High School
Katherine MacKay		Kent Place School
Gaelle Ngabo		Bergen County Academies
Hema Gujjar		Lawrence High School

What we did

Background:

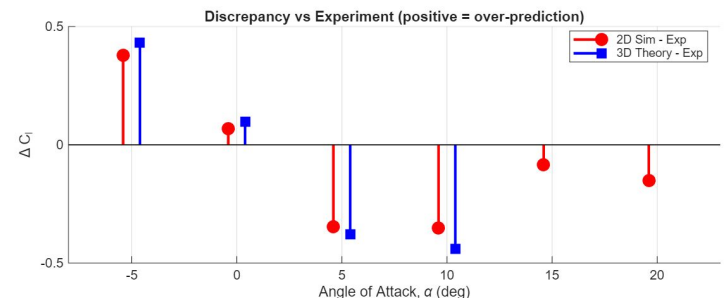
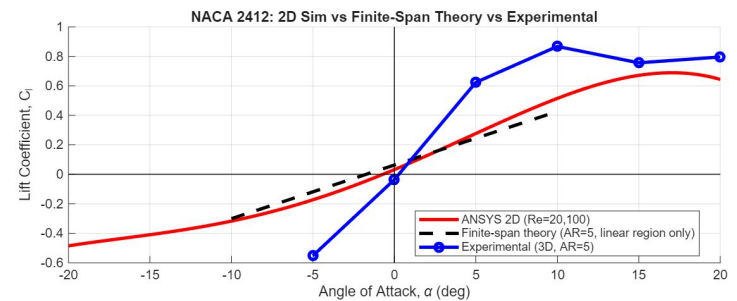
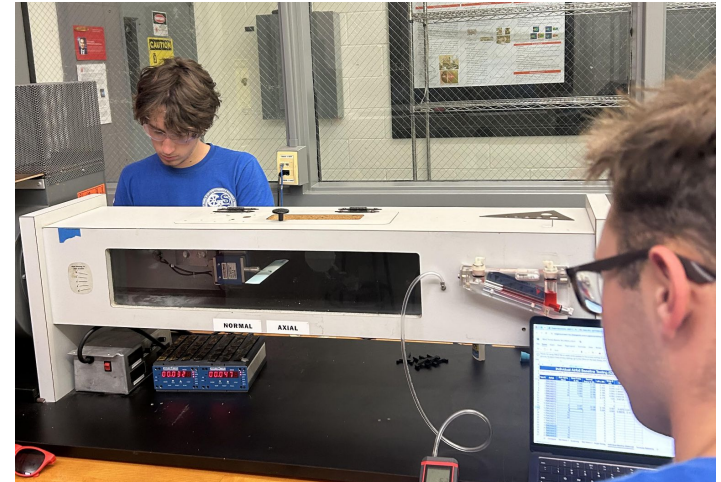
- ❖ **Biplanes** have far less experimental aerodynamic data than monoplanes, despite relevance to short-field, low-speed applications (crop dusting & STOL)
- ❖ **Studied 4 NACA airfoils** in monoplane/biplane configurations
- ❖ **Combined 2D CFD simulation** (XFLR5, ANSYS Fluent) with physical **wind tunnel testing of 3D-printed models** at $Re \approx 20,100$



What we did

Objectives:

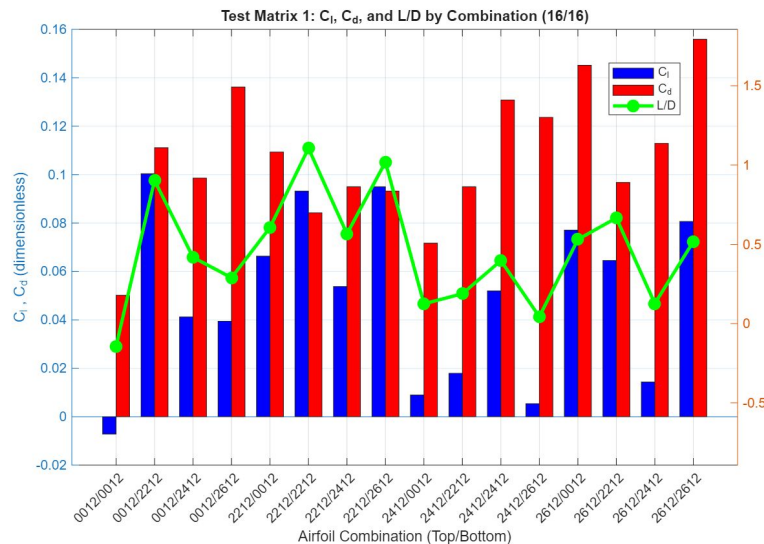
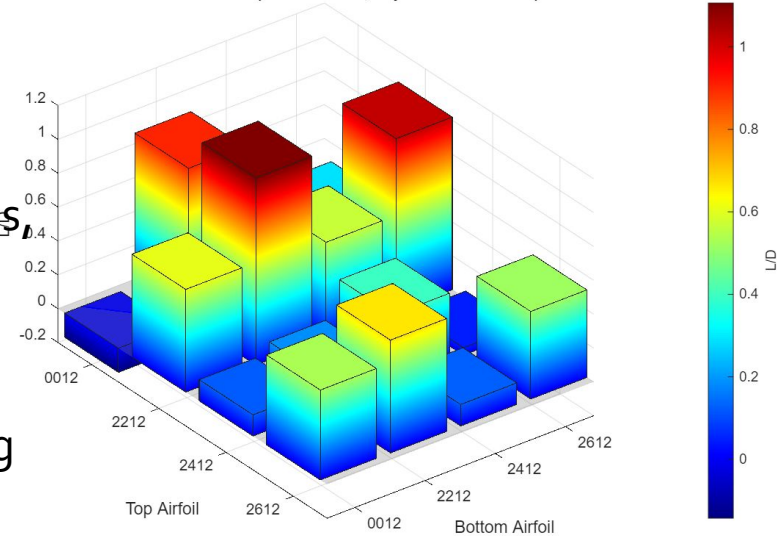
- ❖ Screen 16 possible top/bottom airfoil combinations to identify the most efficient biplane pairing
- ❖ Run a 36-point decalage (angle) sweep on the winning pairing to find the optimal orientation
- ❖ Apply a finite-span/aspect-ratio correction to separate 2D-vs-3D discrepancies from low-Reynolds-number effects when comparing simulation to experiment
- ❖ Complete IEEE-format research paper



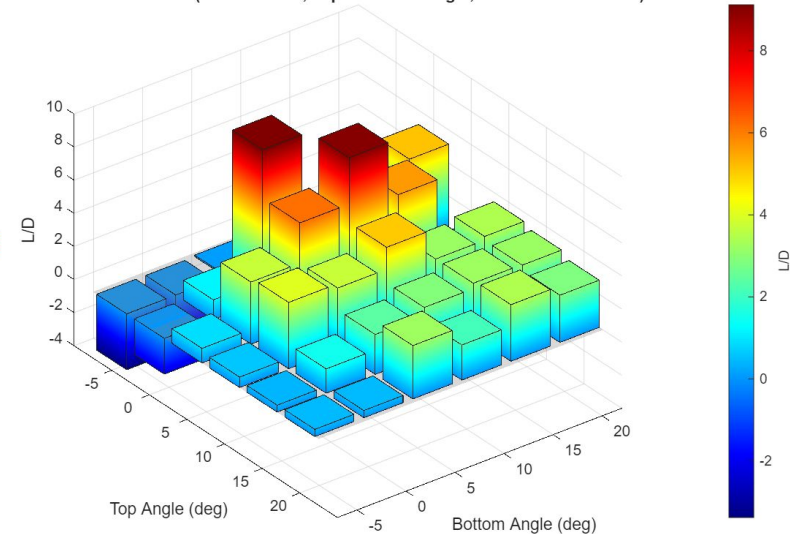
What we found

- ❖ Biplane optimization: $2212 \times 2212 \rightarrow L/D = 1.11$
- ❖ Decalage optimization: $0^\circ \times 5^\circ \rightarrow L/D = 9.12$
- ❖ Best-performing configurations shared a **converging-channel geometry** between the wings, driving a lift-enhancement effect rather than a drag-reduction one
- ❖ Configurations with a **negative bottom-wing angle underperformed**, regardless of the top wing
- ❖ Indicates **lift-dominated** sensitivity

Test Matrix 1: L/D (3D Bar Chart, Top x Bottom Airfoil)



Test Matrix 2: L/D (3D Bar Chart, Top x Bottom Angle, Combo = 2212x2212)



The Benefits

The Mentors:

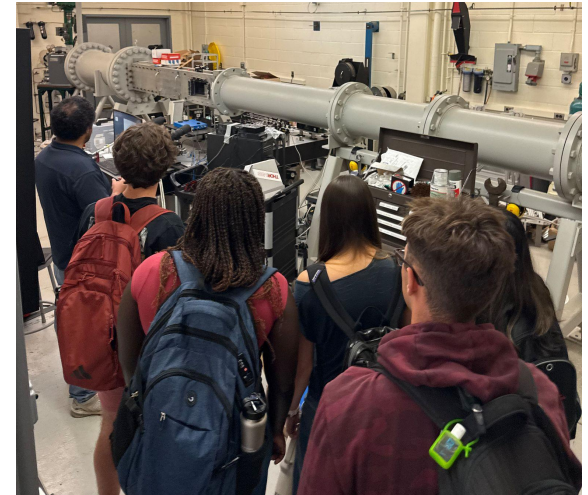
- **Shape the direction and outcome** of a full research project: simulation, testing, and a complete paper
- Manage a team of 5 scholars and shifting priorities
- Develop **communication and leadership skills**

The Students:

- Collegiate level aerodynamics **research experience**
- Hands-on **CFD** (ANSYS Fluent, XFLR5), **wind tunnel testing**, and **CAD/3D printing** (SolidWorks/OnShape)
- Real **engineering exposure**: data collection, error analysis, and technical writing
- **Work directly with Rutgers faculty** (Dr. Edward DeMauro)

Beyond the Cohort:

- Wind tunnel methodology accepted for the Rutgers SOE **Aero/MechE senior lab curriculum**
- **TA that lab**; not yet taken
- Six people this summer → **every senior in two departments**, every year



The project was accepted to the **2026 IEEE MIT Undergraduate Research Technology Conference!**